Electroactive Elastomeric Actuators for the Implementation of a Deformable Spherical Rover

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***Abstract*—This paper presents investigations toward the development of an inflated deformable rolling rover. The proposed novel locomotion system could find potential future use, for example, in scouting and exploration missions. The manufacturing procedure used to fabricate a preliminary prototype is presented in this paper. The prototype consists of a rigid frame and four dielectric elastomer actuator (DEA) sectors, which give the prototype a spherical shape. Analytical models are proposed to predict the electromechanical behavior of the DEA sectors and to estimate their effect on the prototype's dynamics. These models are validated through an experimental procedure.**

***Index Terms*—Concept design, dielectric elastomers, electroactive polymers, rover, smart materials, spherical robot.**

I. Introduction

SPHERICAL rovers have been investigated in recent years [1]–[7] for exploration tasks in different environments, including the outer space. In planetary space explorations, the rover should be lightweight and have low volume in order to reduce the otherwise prohibitive costs of the launch phase. To fulfill these requirements, different innovative propulsion systems for spherical rovers have been developed; for instance, it was proposed that the motion could be generated by an external power source (e.g., wind energy [1]), by the motion of one or more ballasts [2]–[4], or by the deformation of the rover's structure [7].

In this paper, a novel spherical rover, whose external surface deforms upon electrical stimuli, is proposed. The deformation of such an external surface changes the centre of gravity (CG) of the rover, which causes the rover to roll. Differently from previous studies, the external surface consists of an array of dielectric elastomer actuator (DEA) sectors, which can change shape according to the intensity of an imposed electrical field [8]–[11]. The proposed novel spherical rover is inflated to enable prestretching and, therefore, high performance of the DEA array.

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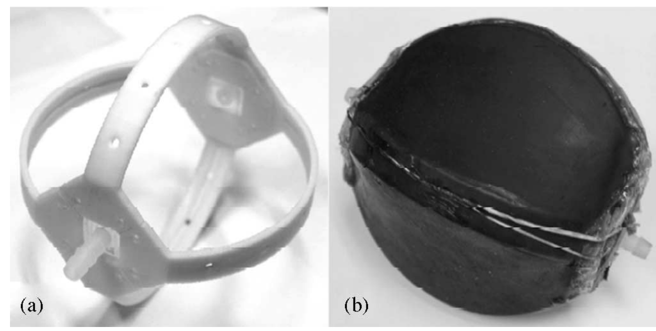


Fig. 1. (a) Internal rigid frame. (b) Prototype (not inflated) covered with DEA sectors.

This paper is organized as follows. Section II introduces the working principle on which the rover is based on and also outlines features that the developed preliminary prototype has. Section III investigates the behavior of the hyperelastic polymer that was used to manufacture the DEAs. Section IV introduces a simplified analytical model developed to predict the dynamic response of the external surface of the prototype; this section also presents an experimental validation of the proposed model. Section V presents both analytical and experimental analyses to estimate the potential rolling performance of the developed prototype. Conclusions of this study are drawn in Section VI.

II. Proposed Concept and Prototype

The proposed spherical system consists of an internal rigid frame (IRF) covered by a dielectric elastomer. The IRF, which was fabricated with a rapid prototyping printer [see Fig. 1(a)], served to confer the system a spherical shape and to support the prototype's weight. The external surface of the prototype was made of a poly(dimethylsiloxane)-based silicone rubber (TC-5005 A/B-C, BJB Enterprises, Inc., Tustin, CA). Following a degassing phase, the polymer was cured in a square Plexiglas mould (30 cm $\times$ 30 cm) having 1.2-mm thickness. Once cured, the polymer was peeled from the mould to obtain a flat strip (15 cm $\times$ 20 cm), which was subsequently wrapped to form a hollow cylinder as proposed in [12], [18]; the hollow polymeric cylinder had diameter smaller than the IRF diameter. The IRF was inserted inside the polymeric cylinder and its lateral extremities were, subsequently, bent inward and bonded with silicone to completely seal the prototype as shown in Fig. 1(b). The particular shape of the IRF geometrically divided the dielectric elastomer in four sectors. The internal surface of the dielectric

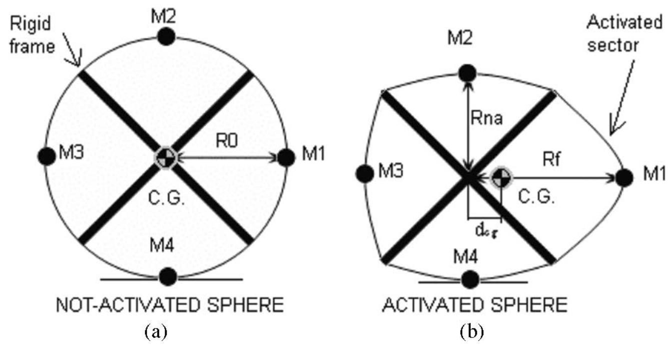


Fig. 2. Activation of one DEA sector and consequent displacement of the prototype CG. (a) Inflated but not electrically activated prototype. (b) Deformation caused by the electrical activation of one DEA sector.

elastomer was covered with a deformable electrode, which was, therefore, common to the four sectors of the elastomer. The external surface was instead covered with four different electrodes to enable independent actuation of each sector, called hereafter DEA sectors. The internal and external electrodes were manufactured by spreading carbon powder on the polymer's surfaces. The prototype was sealed and inflated with air to achieve a spherical shape as schematically represented in Fig. 2(a). Initial conditions of the system, considered for the subsequent analysis, were those in which the prototype had a spherical shape (therefore pressurized) with no electrical field applied. In this configuration, the prototype had internal volume V_0 and initial internal pressure p_0; the system was considered to be isolated and isothermal. In steady-state equilibrium, the pressure p_0 exerted by the pressurized air was balanced by the elastic force of the polymer. Under this condition, the spherical prototype had radius R_0 [see Fig. 2(a)].

In this study, we hypothesized that the electrical activation of each DEA sector mainly changed the stiffness of that sector. Specifically, it was assumed that when voltage was applied, the stiffness of the activated sector decreased, and therefore, it deformed under the load provided by the pressurized air inside the prototype. This hypothesis, experimentally validated in the following sections of the paper, was used to simplify the dynamic analytical model of the actuator.

The motion of the prototype was generated by the displacement of its CG caused by the activation of one or multiple DEA sectors. Fig. 2 schematically represents the proposed working principle. Each DEA sector is represented by a lumped mass M_i. If one sector is electrically activated, its stiffness decreases and the internal pressure deforms it outward thus increasing its radius (e.g., see M_1 and R_f in Fig. 2(b). As the volume changes, the internal pressure decreases. The stiffness of the three inactivated DEA sectors remains, however, unchanged and therefore, due to a drop in internal pressure, their radii decrease until a new equilibrium position is reached (e.g., see M_2–M_4 and R_{na} in Fig. 2). If one sector is activated, the CG of the prototype, therefore, moves toward the electrically activated sector [see the displacement d_{cg} in Fig. 2(b)] and the prototype starts rolling.

III. Mechanical Behavior of the Dielectric Elastomer

In this section, different models are investigated to represent the mechanical properties of the dielectric elastomer used in this study: TC-5005 A/B-C (BJB Enterprises, Inc., Tustin, CA). Specifically, the Neo–Hookean [13], the Mooney–Rivlin [14], and the Ogden models [15] are investigated. These models are all based on the hypothesis that the elastomer is incompressible; the following equations, therefore, hold true:

$$V_0 = V_f \tag{1}$$

$$l_0 w_0 t_0 = l_f w_f t_f \tag{2}$$

where l is the length, w is the width, t is the thickness of the sample, and the subscripts 0 and f are, respectively, the initial and final states. By dividing (2) by V_0, the incompressibility condition can be expressed as follows:

$$\frac{l_f}{l_0} \cdot \frac{w_f}{w_0} \cdot \frac{t_f}{t_0} = a_1 \cdot a_2 \cdot a_3 = 1 \tag{3}$$

where a_1 is the stretch in the direction of the length, a_2 is the stretch in the direction of the width, and a_3 is the stretch in the direction of the thickness of an elastomeric sample. In the three considered hyperelastic models, the stress σ along the i direction is obtained as the derivative of the strain energy function W with respect to the extension ratio

$$\sigma_i = a_i \frac{\partial W}{\partial a_i} - p_h \tag{4}$$

where p_h is a component of an assumed external hydrostatic pressure acting on the sample [15].

In the Neo–Hookean model [13], a linear relation between stress and strain is used in replacement of the linear relationship between force and deformation of the Hooke's model. It is assumed that in a uniaxial tensile test the sample is deformed along one direction, for instance direction 1, and the total stress is a function of the stretch along the same direction, namely,

$$\sigma_1 = G \cdot a_1^2 - \frac{G}{a_1} = G\left(a_1^2 - \frac{1}{a_1}\right) \tag{5}$$

where G is called the shear modulus. The force to deform the sample is the total stress in direction 1 multiplied by the effective cross-section area of the sample [13]

$$f_1 = w_f t_f G \cdot a_1^{-1}\left(a_1^2 - \frac{1}{a_1}\right) = w_0 t_0 G\left(a_1 - \frac{1}{a_1^2}\right). \tag{6}$$

The Mooney–Rivlin [14] solid model is a generalization of the Neo–Hookean solid model. It is assumed that in a uniaxial tensile test the total stress along the direction of deformation is

$$\sigma_1 = \left(2C_1 - \frac{2C_2}{a_1}\right)\left(a_1^2 - \frac{1}{a_1}\right) \tag{7}$$

where C_1 and C_2 are two coefficients experimentally determined. The force necessary to deform the sample is

$$f_1 = w_0 t_0 \left(2C_1 - \frac{2C_2}{a_1}\right)\left(a_1 - \frac{1}{a_1^2}\right). \tag{8}$$

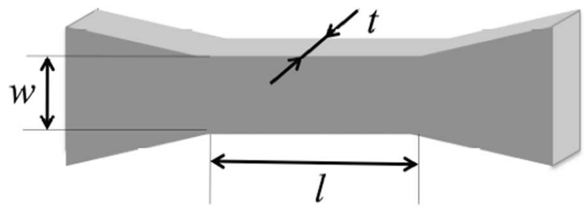


Fig. 3. Bone shape of the sample tested.

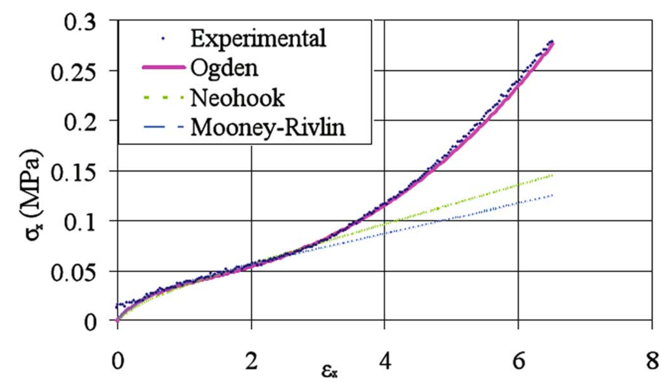


Fig. 4. Experimental TC-5005 stress–strain curve fitted by the Ogden, Mooney–Rivlin, and Neo–Hookean models.

TABLE I
COEFFICIENTS OF HYPERELASTIC MODELS

MODEL	COEFFICIENTS	
Neo-Hookean	$G = 19.3$ kPa	
Mooney-Rivlin	$C_1 = 7.44$ kPa $C_2 = -6.35$ kPa	
Ogden	$\mu_1 = 165$ kPa	$k_1 = 0.344$
	$\mu_2 = 798$ Pa	$k_2 = 3.88$

The Ogden model [15] is based on a different strain energy function compared to the Mooney–Rivlin model; the stress along direction 1 is

$$\sigma_1 = \sum_n \mu_n a_1^{k_n} - \mu_n a_1^{-1/2k_n} \tag{9}$$

where a sum over n is implied and the coefficients μ_n and k_n should be determined experimentally. The force necessary to deform the sample is

$$f_1 = w_0 t_0 \left(\mu_n a_1^{(k_n - 1)} - \mu_n a_1^{(-1-(1/2)k_n)}\right). \tag{10}$$

The mechanical behavior of TC-5005 was experimentally determined through a tensile test by using an Instron 5848 microtester. The sample tested had a bone shape as represented in 3; the initial dimensions of the sample tested were $l_0 = 9.1$ mm, $w_0 = 3.04$ mm, and $t_0 = 2.18$ mm.

As expected, the tests showed high nonlinearity of the considered elastomer. The experimental data were fitted by using the least-squares method (see Fig. 4) in order to determine the experimental coefficients of the three constitutive models represented by the equations previously introduced. The identified coefficients are summarized in Table I.

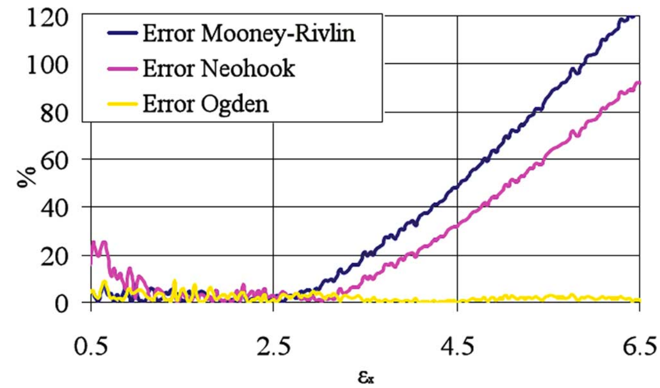


Fig. 5. Error between hyperelastic models and experimental data.

TABLE II
MAXIMUM AND AVERAGE ERROR BETWEEN ANALYTICAL AND EXPERIMENTAL DATA

	Neo-Hookean error (%)	Mooney-Rivlin error (%)	Ogden error (%)
Average	*27.08*	*37.10*	*2.02*
Maximum	*91.98*	*123.60*	*9.33*

An error analysis was performed to identify which hyperelastic model would better represent the behavior of the considered elastomer; the error was calculated as follows:

$$E = \left\| \frac{\sigma_{\text{Exp}} - \sigma_{\text{hyp}}}{\sigma_{\text{hyp}}} \right\| \times 100 \tag{11}$$

where E is the error, σ_{hyp} is the stress predicted by the hyperelastic models, and σ_{Exp} is the measured stress. The obtained results are shown in Fig. 5. The Neo–Hookean model was not very suitable to predict the behavior of TC-5005, as the maximum computed error for the stress was about 92% (see Table II). In addition, this model reported deviation greater than 10% both for strain less than 1 and higher than 3.5 (see Fig. 5. On the other hand, the Mooney–Rivlin model was only suitable to predict relative small strains (0–3 range) as shown in Fig. 5; for strain higher than 3, the error was higher than 10% with a maximum pick of about 124%. The model that better fitted the experimental data was, as expected, the Ogden model. This model, which can be considered a refined version of the Neo–Hookean and Mooney–Rivlin formulations, takes in fact into account also the strain-hardening behavior of the polymer [16]. Table II reports that the Ogden model accurately predicted the experimental data—the maximum error was about 9% and the average error was about 2%. Due to its better performance, the Ogden model was considered in the analytical models presented in the following sections.

IV. DEA SECTORS DYNAMICS

A. Spherical Model

A simplified 3-D lumped parameters model was developed to represent the dynamic behavior of a single DEA sector of the prototype. Each DEA sector was considered rigidly fixed to the

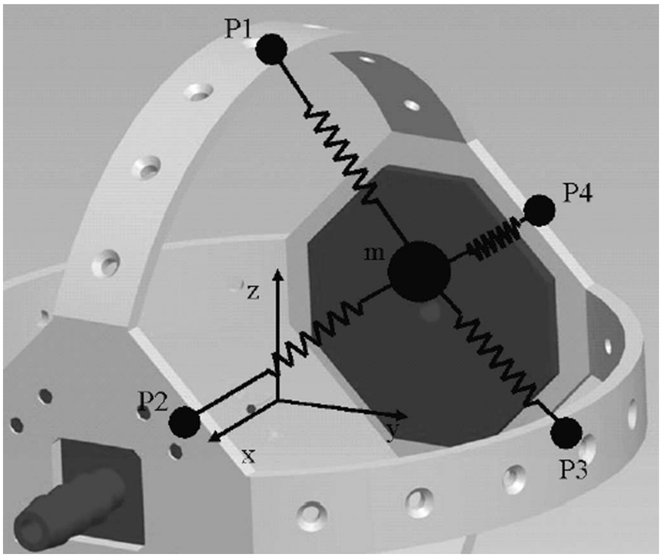


Fig. 6. IRF CAD model and lumped-parameter representation of a DEA sector.

TABLE III
DEFINITION OF SYMBOLS USED IN (12)

Symbol	*Definition*
$F_{k,i}$	elastic force
Fc_i	damping force
k_i	spring stiffness
c_i	damping coefficient
x,y,z	lumped-mass coordinates
x_{Pi}, y_{Pi}, z_{Pi}	coordinates of the fixed point P_i
F	concentrated force generated by the internal pressure
p	internal pressure
R_0	radius of the sphere
v_{Pi}	velocity of the lumped-mass projected along the direction defined by the position of the lumped-mass itself and the point P_i

IRF (as physically implemented in the prototype). The model consisted of a single lumped mass (the mass of the DEA sector) and four symmetrically positioned spring-damper elements connecting the mass to the IRF, as schematically presented in Fig. 6. By using Newton's law, the following equations were obtained:

$$\begin{cases} m\ddot{x} = (Fk_i + Fc_i + F)_x \\ m\ddot{y} = (Fk_i + Fc_i + F)_y \\ m\ddot{z} = (Fk_i + Fc_i + F)_z \\ F_{k,i} = k_i(\sqrt{(x - x_{pi})^2 + (y - y_{pi})^2 + (z - z_{pi})^2} - l_{0i}) \\ Fc_i = c_i \cdot v_{Pi} \\ F = \pi \cdot R_0^2 \cdot p \end{cases} \tag{12}$$

where, parameters are defined in Table III, subscripts *x, y*, and z refer to the axes of a Cartesian reference frame (see Fig. 6), and subscript i refers to the specific considered fixed points (for instance, $F_{k,1}$ is the elastic force between the mass and the fixed point P1).

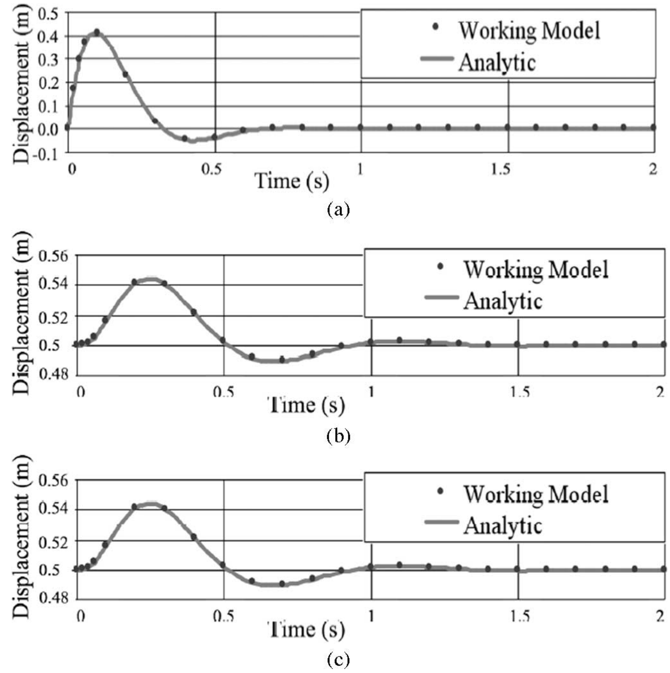


Fig. 7. Numerical validation of (12). (a)–(c) Displacements along axes *x, y*, and z of the lumped mass of one elastomer sector.

Equations (12) were validated by using a commercial multibody dynamics software (working model); different initial conditions were considered and the dynamic response of the model was computed. In the scenario considered for validation, the properties of the elastic elements were assumed to be constant and no external forces were applied to the system. Fig. 7 shows displacements of the lumped mass of one sector in the three orthogonal directions x, y, and z for an exemplificative case in which the lumped mass had an initial velocity equal to $v_x = 10$ m/s. It can be seen in Fig. 7 that our model represented by (12) was consistent with the simulation performed in working model.

The complete analytical model of the spherical prototype was subsequently developed based on (12). The sphere, according to the prototype shown in Fig. 1, had four different sectors and the dynamics of each sector was described by (12). The main additional aspect to be considered was that the deformation of one sector affected the others due to the induced variation of the internal pressure inside the prototype (see Fig. 2). According to Boyle's law, the pressure inside the prototype changes according to

$$p_0 V_0 = p_f V_f. \tag{13}$$

When one sector was activated, the variation of the volume inside the sphere caused a change in pressure, and, as a consequence, also a change in the concentrated force F [see Table III and (12)].

In order to finalize the developed simplified model of the prototype, the effect of the electrical activation of the DEA sectors was taken into account, as discussed in the following section.

B. Model of the Variable DEA Stiffness

The tensile stiffness of TC-5005 is not constant as shown in the experimentally obtained stress–strain curves in Fig. 4. In addition, it should be noted that the DEA tensile stiffness changes when an electrical field is imposed, as a consequence of the induced Maxwell stress [10]–[12]. By using the high-strain model of actuation under load presented in [17], the following relationship was obtained:

$$S + \frac{\varepsilon_{\rm rel}\varepsilon_0 w_0}{t_0} T^2 a_1 = F_k \tag{14}$$

where S is the external load, $\varepsilon_{\rm rel}$ is the relative dielectric permittivity, ε_0 is the absolute dielectric permittivity, T is the voltage applied, and F_k is the elastic force. The elastic force of the material F_k was computed according to the Ogden hyperelastic model (the experimentally obtained coefficients μ_n and k_n presented in Table I were used). In this study, we hypothesized that the width dimension of the dielectric sector did not change under the imposed loading conditions (namely $a_2 = 1$) as proposed in [16]. The following equation was, therefore, used to model the elastic force:

$$F_k = w_0 t_0 \sum_n \mu_n \left(a_1^{(k_n - 1)} - a_1^{(-1-k_n)}\right). \tag{15}$$

Starting from (14) and applying the definition of tensile stiffness, an analytical model of the hyperelastic stiffness of the material was obtained

$$k_i = \frac{S}{\Delta l} = \frac{F_k - (\varepsilon_{\rm rel}\varepsilon_0 w_i / t_i T^2) a_1}{l_f - l_0} \tag{16}$$

where Δl is the elongation of the sample in the direction of the applied external load. Equations (12), (13), and (16) describe the dynamic model of the actuation of the sphere.

C. Parameters Used in Numerical Simulations

The parameters in (12), (13), and (16) were experimentally obtained from the prototype shown in Fig. 1. The stiffness of each spring, according to (16), depended on the dimension of the spring itself, namely on its width (w_i), length (l_i), and thickness (t). The thickness of each DEA sector was related to the thickness of the mould used to fabricate the elastomer while the width and length of each spring depended on the specific IRF geometry. These two latter parameters were computed such that the area defined by w_i and l_i was the same to the area approximately identified by connecting the points P1, A, P2, P3, and B shown in Fig. 8. As the polymer covering the IRF was approximately prestretched about 1.3%, a prestretching coefficient (δ) equal to 1.3 was used to compute the initial relaxed length and width of each spring. The experimentally obtained parameters are summarized in Table IV.

In order to validate the model, different tests were performed. Only quasistatic experiments were carried out—dynamic tests were considered to be not strictly needed in this study, which mainly aimed at presenting the feasibility of the proposed new concept. Extensive dynamic tests will be performed in the future and reported in a future independent article. Electrical activation and inflating tests are presented in the following sections.

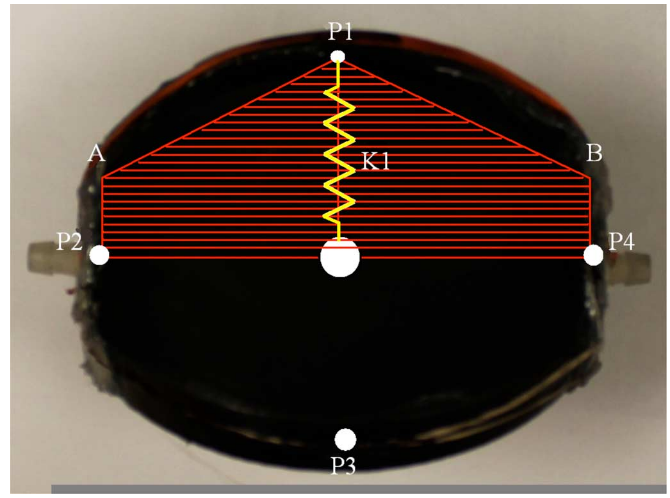


Fig. 8. Front view of a sector.

TABLE IV
PARAMETERS USED TO SIMULATE THE ACTUATION OF THE SECTORS

Parameter	Unit	Value
R_0	mm	53
W_1	mm	54.8
w_2	mm	36.5
w_3	mm	54.8
w_4	mm	36.5
L_1	mm	18.3
l_2	mm	27.4
l_3	mm	18.3
l_4	mm	27.4
t	mm	1.2
ε_{rel}	-	10
ε_0	Fm^{-1}	$8.85 \cdot 10^{-12}$
M	g	10
P	Pa	572
δ	-	1.3

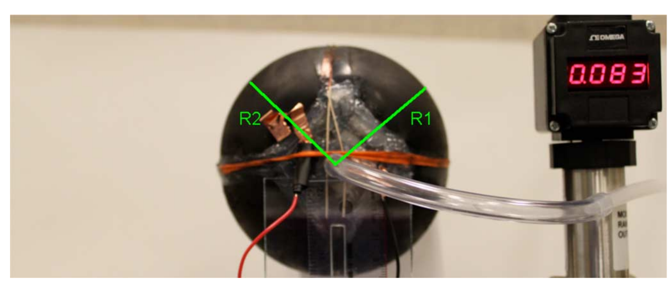


Fig. 9. Side view of the inflated sphere.

D. Tests

The inflating test consisted on pressurizing with air the prototype until a spherical shape was reached (see Fig. 9). The internal

TABLE V
EXPERIMENTAL AND ANALYTICAL VALUES OF TWO RADII OF THE SPHERE WHEN INFLATED WITH A RELATIVE PRESSURE OF 576 Pa

R1 analytical (mm)	R2 analytical (mm)	R1 exper. (mm)	R2 exper. (mm)	Error R1 (%)	Error R2 (%)
54.54	*54.54*	*54.17*	*56.05*	*0.68*	*2.77*

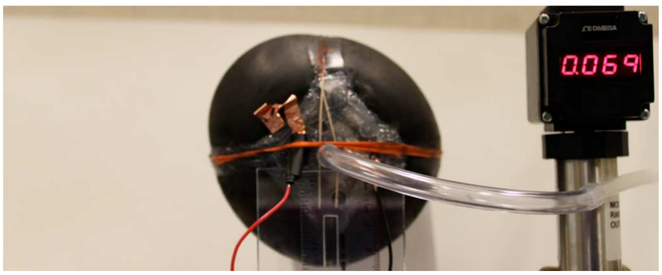


Fig. 10. Inflated prototype with one sector electrically activated (electrical field applied was about 9.3 V/μm).

pressure was measured by a pressure sensor (Omega PX4200–010GI); the external radius of the sphere was instead identified by postprocessing images (Vision Builder software, National Instruments) of the inflated prototype. A spherical shape was obtained when the internal relative pressure reached about 576 Pa. The deviation of the analytical prediction from the experimental data was computed as

$$E_r = \left\| \frac{R_{\exp} - R_{\text{analytical}}}{R_{\text{analytical}}} \right\| \times 100 \quad (17)$$

where E_r is the error between the experimentally measured ($R_{\exp}$) and the analytically predicted ($R_{\text{analytical}}$) radii. Table V presents results obtained during a test. R_1 and R_2 are the radii of two DEA sectors as shown in Fig. 9. Discrepancy between R_1 and R_2 is due, mainly, to the manual manufacturing procedure used to fabricate the prototype. It can be seen that the analytical and experimental results are consistent.

Once inflated, electrical activation tests were performed. Starting from an inflated sphere, the DEA sector #2, corresponding to R_2 (see Fig. 9), was electrically activated. As expected, the activated sector moved outward (R_2 increased) whereas the other three sectors moved inward (for example R_1 decreased). Fig. 10 presents the case in which an electrical field equal to 9.3 V/μm was applied to the DEA sector #2.

Fig. 11 shows both experimental results and analytical predictions for R_1 and R_2 when different electrical fields were applied. It can be seen that the analytical model takes well into account the variation in Maxwell pressure due to the electrical field, the nonlinear behavior of the dielectric elastomer, and also the variation of the internal pressure due to the variation of the volume of the prototype, which all affect the variation of the radii. Table VI confirms this statement: the maximum recorded E_r was less than 3%, which is very low when taking into account the limitations of the manufactured prototype.

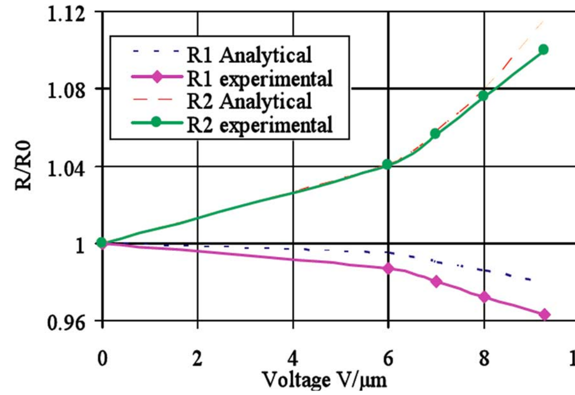


Fig. 11. Relative variations of R_1 and R_2 recorded when the DEA sector #2 was electrically activated.

TABLE VI
EXPERIMENTAL AND ANALYTICAL ACTIVE (R_2) AND INACTIVE (R_1) RADII OF THE SPHERE

Volt (V/μm)	R1 analyt. (mm)	R2 analyt. (mm)	R1 exper. (mm)	R2 exper. (mm)	Error R1 (%)	Error R2 (%)
0	*54.54*	*54.54*	*56.11*	*54.38*	*2.87*	*0.29*
6	*54.26*	*56.78*	*55.39*	*56.57*	*2.27*	*0.37*
7	*53.99*	*57.71*	*54.99*	*57.44*	*1.85*	*0.46*
8	*53.77*	*58.88*	*54.55*	*58.5*	*1.45*	*0.64*
9.3	*53.40*	*60.83*	*54.04*	*59.79*	*1.20*	*1.71*

TABLE VII
DEFINITION OF THE PARAMETERS USED IN (18)

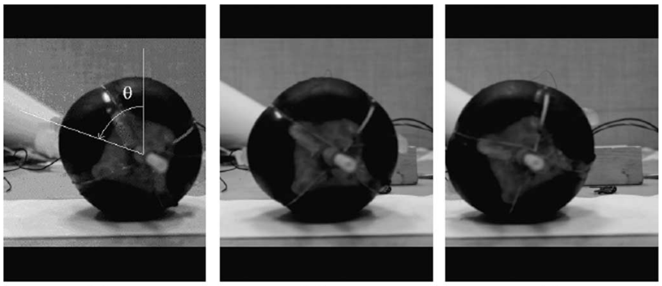
Prototype total mass, m_s (g)	*92*
Frame mass, M (g)	*40*
Mass of one sector, m (g)	*13*
Initial radius of each sector, R_0 (mm)	*54*
Inertia of the prototype, I_s (kg·m^2)	*1.79·10^{-3}*

V. SPHERE ROTATION

The prototype had a spherical shape and its revolute motion was generated by the displacement of the CG caused by the activation of one or multiple DEA sectors. The following simplified single lumped mass model of the sphere was used to predict the rotation of the prototype:

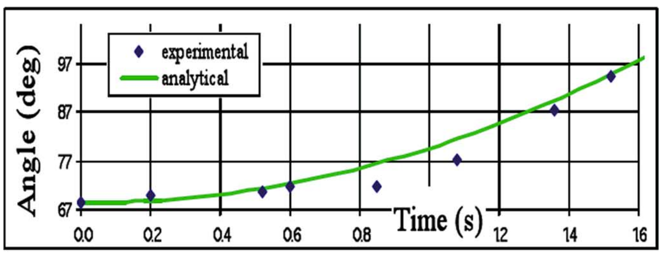
$$I_s \times \ddot{\theta} = (m_s \times g \times d_{\text{cg}}) \times \sin\theta \quad (18)$$

where I_s is the inertia of the prototype, which was assumed to be constant and was computed starting from the CAD model of the prototype, θ is the angle describing the rotation of the sphere, m_s is the mass of the sphere, g is the gravitational acceleration, and d_{cg} is the prototype's CG displacement caused by the activation of the DEA sectors; sin(θ) was used to take into consideration the torque variations caused by the rotation of the prototype. It should be noted that the deformations of the DEA sectors were computed according to (12), (13), and (16). The parameters used in the analytical model are reported in Table VII. Fig. 12 shows different snapshots of the rolling prototype.

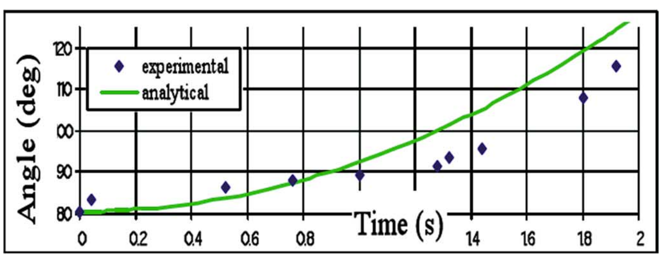


(a) (b) (c)

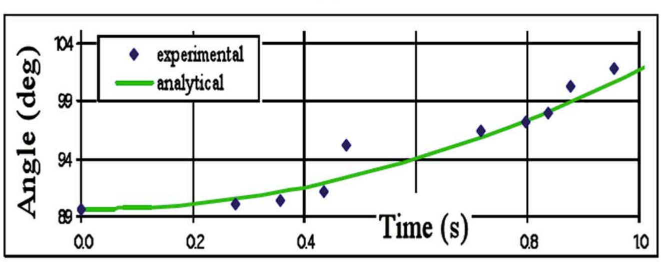
Fig. 12. Snapshots during a rotation test of the prototype.



(a)



(b)



(c)

Fig. 13. Comparison between experimental and analytical data for the manufactured prototype. (a) Initial angle was 68.3°. (b) Initial angle was 80.5°. (c) Initial angle was 89.6°.

Fig. 13 presents results about three tests in which the prototype had three different initial positions [the initial position is identified by the rotation angle θ shown in Fig. 12 (a)]. In the three tests reported in Fig. 13, a single DEA sector was activated. For each test, a video was recorded and its frames analyzed to determine the prototype's performance. Fig. 13 shows that the analytical prediction was consistent with the experimental results. Table VIII reports results of the error analysis that was performed. It can be seen that the maximum deviation of the analytical prediction from the experimental results was, on average, about 8%, which is relatively low considering the imperfections that the manually fabricated prototype had.

TABLE VIII
MAXIMUM AND AVERAGE ERROR BETWEEN THE ANALYTICAL AND EXPERIMENTAL DATA IN FIG. 13

	Exper. 1 Fig. 13(a)	Exper. 2 Fig. 13(b)	Exper. 3 Fig. 13(c)	Average
	Error (%)	Error (%)	Error (%)	Error (%)
Max	11.8	9.3	3.0	8.0
Average	3.2	5.2	0.8	3.1

Fig. 13 allows determining the potential performance of the prototype. Specifically, a maximum angular velocity of 0.8 rad/s was reached, corresponding to a linear velocity of 43 mm/s for our prototype, which had an external radius equal to 54 mm. The recorded maximum angular acceleration was 0.42 rad/s^2 (22 mm/s^2 linear velocity for our prototype).

VI. CONCLUSION

This paper presented and validated a new conceptual design for future lightweight rolling rovers potentially suitable for exploration and scouting missions. The proposed prototype consisted of an IRF and an external deformable surface made of DEA sectors. Different analytical models were developed to investigate the behavior of the DEAs, the dynamics of each DEA sector, and the rolling performance of the prototype. Specifically, a comparison among different hyperelastic models was performed to represent the behavior of the used dielectric elastomer; it was shown that the Ogden model could describe the stress–strain curve with a maximum error of 9%. Experimental tests were performed to validate a 3-D model developed to predict radial displacements of the sectors of the prototype; the maximum deviation between analytical and experimental results was 3%. Experimental tests were also performed to identify the rolling performance of the prototype; the proposed analytical model was able to describe the rotation angle of the prototype with a maximum error of 11.8%. The maximum angular acceleration of the developed prototype was 0.42 rad/s^2. The fabricated prototype could potentially fulfill stringent requirements concerning mass and volume; indeed, the whole weight of the prototype was only 75 g and its volume, once inflated, was 660 cm^3—the prototype could potentially have a total folded volume (in the case of foldable internal frame) of about 30 cm^3.

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